

LITERATURE SURVEY

ContextGraph

A Cross-AI Personal Context Engine

To situate ContextGraph, existing approaches to persistent AI context and memory were reviewed across three groups: memory features built into commercial chat products, open agent-memory frameworks and research, and protocol-level or knowledge-graph approaches to portable context. A summary comparison is given first, followed by a short discussion of each group and the resulting research gap.

1. Comparative Summary

#	System / Work	Year	Key Idea	Limitation (relative to this project)
1	ChatGPT Memory (OpenAI)	2024	Model silently extracts and stores user facts across conversations.	Single flat store, one vendor only, no explicit hierarchy or user-triggered save.
2	Claude Projects / Memory (Anthropic)	2025	Project-scoped knowledge plus persistent memory inside Claude apps.	Scoped to Claude only, not reachable by other AI tools, no cross-project graph.
3	Model Context Protocol (Anthropic)	2024	Open JSON-RPC standard letting any AI client call external tools/resources.	Defines the transport/tool-call standard only, not a memory data model.
4	MemGPT (Packer et al., arXiv:2310.08560)	2023	Treats the LLM context window like OS virtual memory, paging facts between core and archival tiers.	A single-agent runtime construct, not a hosted multi-user, multi-client service.
5	Mem0 (arXiv:2504.19413)	2025	Extracts, scores, and retrieves memories for agents with a production-oriented pipeline.	A memory SDK/library, with no scoped-hierarchy model or MCP-native distribution built in.
6	Zep (Rasmussen et al., arXiv:2501.13956)	2025	Temporal knowledge graph that tracks how facts about a user change over time.	Strong on temporal graph reasoning, not built around per-user API keys or a visual dashboard.
7	LangChain / LangGraph memory modules	2024	Composable memory abstractions (buffer, summary, vector-store backed) for custom agents.	A toolkit for assembling a bespoke memory system, not an out-of-the-box product.
8	Graph-based PKM tools (Obsidian, Roam)	N/A	Let a person manually build and browse a linked-note knowledge graph.	Manual curation only, no AI read/write access, and no relevance decay.
9	Generative Agents (Park et al., arXiv:2304.03442)	2023	Gives a simulated agent a memory stream, retrieved by recency, importance, and relevance, with periodic reflection into higher-level notes.	Memory lives inside one simulated agent process, not exposed through a protocol multiple external AI tools can share.

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10	GraphRAG (Edge et al., Microsoft, arXiv:2404.16130)	2024	Builds a knowledge graph over a document corpus, with community summaries for query-focused retrieval.	Built for corpus-level document question answering, not a small, continuously-updated, per-user context store.
11	Retrieval-Augmented Generation (Lewis et al., arXiv:2005.11401)	2020	Foundational technique pairing a parametric language model with a non-parametric retrieval index.	A general retrieval mechanism; says nothing about how personal context should be scoped, judged, or organized as a hierarchy.
12	A-Mem (arXiv:2502.12110)	2025	Lets an agent structure, link, and evolve its own memory notes, inspired by the Zettelkasten method.	Memory organization is internal to one agent's note graph, not an externally addressable, multi-client, scope-aware store.

2. Commercial Memory Features

ChatGPT Memory and Claude's Projects/memory both improved on stateless chat by letting the model retain facts across sessions. Both remain single-vendor: context built up inside one product cannot be read by, or moved to, a different AI tool, which is precisely the situation ContextGraph users are in, since they routinely use more than one assistant.

3. Open Agent-Memory Frameworks

MemGPT, Mem0, and Zep each treat memory as an engineering problem inside a single agent runtime: how to page, score, or temporally index facts so an LLM with a bounded context window can still reason over a long history. They are memory *engines* that a developer wires into their own agent, useful building blocks, but none ship as a hosted, per-user service with an API key and a dashboard a non-developer end user can look at.

4. Protocol and Knowledge-Graph Approaches

The Model Context Protocol (MCP) solves a different half of the problem: it standardizes how an AI client calls an external tool or reads an external resource, regardless of which AI vendor built the client. It says nothing about what shape that external memory should take. Graph-based personal knowledge management tools (Obsidian, Roam) show that a linked-node graph is a workable mental model for personal context, but they are built for manual human browsing, not for being queried and updated by an AI agent mid-conversation.

5. Academic Foundations

Beyond commercial products and open frameworks, three lines of academic research shaped how ContextGraph's data model was designed.

- **Generative Agents** (Park et al., 2023) gave a simulated agent a *memory stream*, scored by recency, importance, and relevance, and periodically compressed into higher-level reflections. ContextGraph borrows the scoring idea (its own relevance score decays weekly) but replaces automatic reflection with an explicit, user-triggered judgment step, so a save is traceable to a reason rather than an opaque model decision.

- **GraphRAG** (Edge et al., Microsoft Research, 2024) showed that structuring retrieval as a graph, rather than a flat vector index, improves multi-hop, corpus-wide questions. ContextGraph applies the same insight at a much smaller scale: a per-user graph of scoped nodes and edges, built for continuous personal use rather than one-off document corpora.
- **Retrieval-Augmented Generation** (Lewis et al., 2020) established the general pattern of pairing an LLM with an external index so it is not limited to what was in its training data. ContextGraph is a specialized case of this pattern where the index is a small, structured, user-owned graph instead of a large document store.
- **A-Mem** (2025) let an agent write and link its own memory notes in a Zettelkasten style, so structure emerges from the agent itself rather than a fixed schema. ContextGraph takes the opposite, more predictable trade-off for a personal-productivity tool: a fixed node and edge schema (scope, tags, relevance) that a user can read and edit directly, rather than an agent-authored structure only the model fully understands.

6. Research Gap

No reviewed system, commercial, open-source, or academic, combines all of the following: (i) a hierarchical, scope-aware context model that prevents cross-project leakage, (ii) access through a standard, multi-client protocol rather than a single vendor's app or a single agent's runtime, (iii) an explicit, user-visible save and relevance mechanism instead of silent inference, and (iv) a consumer-facing visual graph the user can inspect and edit directly. ContextGraph combines these four properties (hierarchy, protocol-level portability, explicit judgment, and visualization) into a single hosted service.

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